KAIHUAI DENG

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EDUCATION

Nanjing University of Information Science and Technology (NUIST) ***Nanjing, China***

B.S. in Atmospheric Science (Hydrometeorology) ***Sep 2023 - Jul 2027 (Expected)***

NUIST ranked #1 in China and #8 globally for Atmospheric Science (2026 ShanghaiRanking GRAS).

- Weighted average: 85.53/100 | GPA: 3.57/4.0 | Class rank: 6/51 **(top 12%)**
- Relevant coursework: Probability & Statistics, Remote Sensing for Hydrometeorology, Hydrometeorological Modeling, Numerical Weather Prediction, Python Programming.

RESEARCH INTERESTS

Remote Sensing · Hydroclimate Extremes · Land–Atmosphere Interactions · Climate Risk Analysis

PUBLICATIONS & MANUSCRIPTS

[1] **Deng, K.**, Huang, Z.* *Area Expansion Dominates Rising Global Three-Dimensional Terrestrial Water Storage Drought Severity.* Under review at *Geophysical Research Letters.*

[2] **Deng, K.**, Huang, Z.*, Li, H., Wu, J. *Global Unequal Exposure to Intensifying Hydrological Droughts.* R4 revision submitted to *Journal of Hydrology.*

[3] Wang, J.*, **Deng, K.** (2024). *A Model that Explains the Contrasting SST Trends in the Southern Pacific Ocean. Science and Technology of Engineering, Chemistry and Environmental Protection.* https://doi.org/10.61173/xnag2k18

** Corresponding author.*

CONFERENCE PRESENTATIONS

Deng, K., Huang, Z. (2026). *Area expansion dominates rising global three-dimensional terrestrial water storage drought severity.* AGU26, San Francisco, USA, 11 December 2026. Accepted poster; presenting author.

Global 3D TWS drought research. English research presentation, National Outstanding Undergraduate Forum, Department of Earth System Science, Tsinghua University, June 2026. (70 participants nationwide)

RESEARCH EXPERIENCE

Mitacs Globalink Research Internship ***Charlottetown, PE, Canada***

Research intern | Advisor: Prof. Xander Wang (University of Prince Edward Island) ***Jul 2026 - Oct 2026***

- Integrated shoreline epochs (1968–2022) and coastal protection records from the PEI Climate Hazard & Risk Information System in QGIS; calculated change statistics weighted by length and data coverage by coastal setting.
- Compared historical and recent erosion hotspots using 1,052 fixed transects in 51 spatial groups at Naufrage; found that three of 11 recent hotspots overlapped historical hotspots.
- Presented findings in lab meetings and coauthored an interim research report.

Global 3D Terrestrial Water Storage Drought Tracking ***Sep 2025 - Present***

Research assistant | Advisor: Prof. Zhongwei Huang (NUIST)

Key Laboratory of Hydrometeorological Disaster Mechanism and Warning

- Built a global drought tracking algorithm for GRACE/GRACE-FO observations (2002–2025) using EEMD, GLDAS-2.2 CLSM gap filling, eight-neighbor spatial connectivity, and 16% directional overlap between months.
- Reconstructed 672 drought events, including splits and mergers; tested overlap thresholds of 10–22% and compared event precipitation and actual evapotranspiration anomalies from ERA5-Land.
- Produced an event catalog and NetCDF event labels and quantified first-order area and intensity effects on severity; led algorithm implementation, sensitivity analysis, visualization, and manuscript preparation.

Global Unequal Exposure to Intensifying Hydrological Droughts ***Jun 2025 - Oct 2026***

Research assistant | Advisor: Prof. Zhongwei Huang (NUIST)

Key Laboratory of Hydrometeorological Disaster Mechanism and Warning

- Applied EEMD to GRACE/GRACE-FO observations, calculated drought frequency, duration, and intensity, and combined drought indicators with GPW population grids; assessed trends using Theil–Sen estimation.
- Estimated that the population exposed to drought increased by 38.76 million people per year; separated hydroclimatic (73.2%), population (5.3%), and interaction (21.5%) contributions.

- Led study design, global data processing, visualization, and manuscript drafting; contributed to revision analyses examining mission gaps and population assumptions.

Fluctuation Theorem & TCWV Asymmetry ***Sep 2024 - Sep 2025***

Research assistant | Advisor: Prof. Jun Yin (NUIST)

Key Laboratory of Hydrometeorological Disaster Mechanism and Warning

- Investigated moisture fluctuations and their relationship with precipitation using monthly ERA5 data (1940–2024); constructed probability density functions and calculated asymmetry A_{asym} using moving averages over different years.
- Estimated an SST–TCWV regression slope of 0.0645 K^{-1} ($R^2 = 0.95$); compared positive and negative distribution tails and conditional relationships among moisture, vertical motion, and precipitation.
- Independently developed the analyses, figures, and manuscript draft; discussed the framework with Prof. Lamberto Rondoni (Politecnico di Torino).

WRF Precipitation Sensitivity over Jing-Jin-Ji ***Mar 2025 - Jul 2025***

Advisor: Dr. Mengya Wang (NUIST)

- Configured WRF/WPS on Linux HPC for 5–13 July 2021 with 27/9/3 km nested domains; used WSM6, YSU, and KF as the baseline and disabled cumulus parameterization in the 3 km domain.
- Compared precipitation responses across physics configurations using spatial correlation, RMSE, and threat scores; assessed differences across rainfall processes and intensity classes.
- Independently prepared model inputs, configured namelist files, managed HPC jobs, and completed the experiment report; used ndown for nested simulations.

Precipitation–Runoff Prediction in the Yiluo River Basin ***Jun 2024 - Mar 2025***

Research assistant | Advisor: Prof. Xing Yuan (Institute of Atmospheric Physics, Chinese Academy of Sciences)

- Applied EOF to CN05.1 monthly precipitation interannual increments to characterize variability between years; screened SST and soil moisture predictors using correlation analysis and stepwise regression.
- Contributed to BiLSTM precipitation prediction and CSSPv2 streamflow experiments driven by precipitation forecasts; evaluated monthly streamflow forecasts and supported model execution on HPC.
- Team results: three PCs explained >90% of variance; precipitation TCC improved by 16% and RMSE decreased by 20% relative to statistical downscaling; monthly streamflow NSE reached 0.65.

Contrasting SST Trends in the Southern Pacific Ocean ***Mar 2024 - Sep 2024***

Student researcher | Advisor: Prof. George Tselioudis (Columbia University / NASA GISS)

- Analyzed Copernicus SST, DUACS sea surface height, CERES radiation, NCEP winds, and NSIDC sea ice across the Southern Pacific (1993–2021); calculated spatial trends and statistical significance.
- Derived Ekman transport and pumping from wind stress and compared sea ice and radiative changes to develop a conceptual explanation of contrasting SST trends.
- Collaborated on literature review, data analysis, conceptual modeling, and manuscript writing; second author of the resulting publication.

INTERNSHIP

Leizhou Meteorological Bureau, Guangdong ***Jul 2025 - Aug 2025***

Intern | Forecast Division ***Zhanjiang, China***

- Supported operational monitoring and warning for Typhoon Danas (2025) and Typhoon Wipha (2025), tracking intensification and landfall trajectories and comparing ECMWF, CMA, NCEP, and TRAMS forecasts.
- Participated in calibration and telemetry checks for meteorological field instruments.

SELECTED AWARDS & HONORS

Provincial Undergraduate Training Program: Principal Investigator (provincial-level project) *May 2026*

Mitacs Globalink Research Internship Award: CSC-Mitacs Joint Program (≤260 awardees nationwide) *Jan 2026*

Second Class Academic Scholarship: NUIST (top 20%) *Nov 2024, Nov 2025, Nov 2026*

Merit Student: NUIST (top 15%) *Dec 2024, Dec 2025, Dec 2026*

ADDITIONAL INFORMATION

Programming: Python, Shell scripting, Fortran, Typst/LaTeX

Scientific Computing: Google Earth Engine, Linux/HPC, Git, WRF/WPS, MATLAB, ArcGIS, QGIS

Statistical & Machine Learning Methods: TensorFlow, PyTorch, BiLSTM, EOF, EEMD

Languages: Mandarin (native), Cantonese (native), English (proficient), French (basic)

Academic Service: Creator and maintainer of othodeng.github.io, publishing study notes from 10+ courses